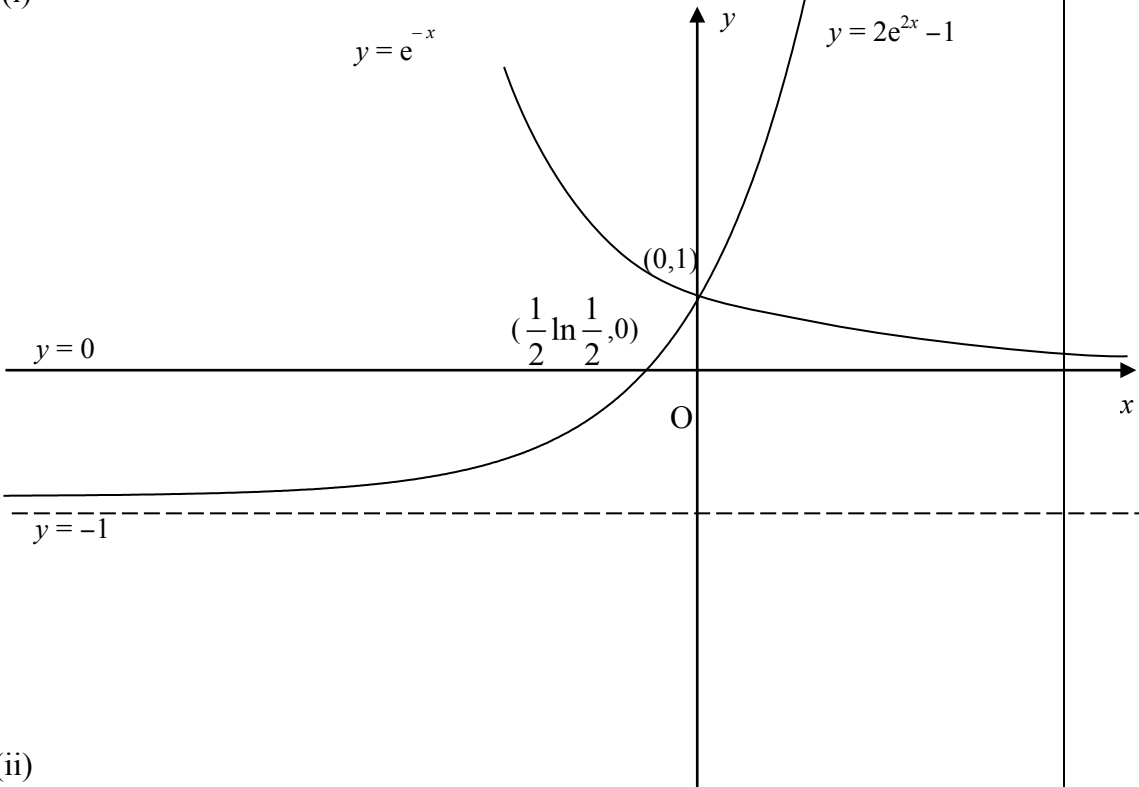


Section A: Pure Mathematics										
1	(i) $15 = 20 - 15e^{-0.01t}$ $5 = 15e^{-0.01t}$ $e^{-0.01t} = \frac{1}{3}$ $-0.01t = \ln \frac{1}{3}$ $t = 109.86 = 110 \text{ (3 s.f.)}.$ (ii) As $t \rightarrow \infty$, $e^{-0.01t} \rightarrow 0$. Therefore $\theta = 20$.									
2	$x + y = 100$ $P = xy = x(100 - x) = 100x - x^2$ $\frac{dP}{dx} = 100 - 2x$ When $\frac{dP}{dx} = 0$, $100 - 2x = 0 \Rightarrow x = 50, y = 50.$ Method 1: Using the first derivative test, <table border="1"><tr><td>x</td><td>50^-</td><td>50</td><td>50^+</td></tr><tr><td>Sign of $\frac{dP}{dx}$</td><td>+</td><td>0</td><td>-</td></tr></table> P is maximum when $x = 50, y = 50$. Method 2: $\frac{d^2P}{dx^2} = -2 < 0.$ Therefore, P is maximum when $x = 50, y = 50$.	x	50^-	50	50^+	Sign of $\frac{dP}{dx}$	+	0	-	
x	50^-	50	50^+							
Sign of $\frac{dP}{dx}$	+	0	-							
3	(a)									

	$\int \frac{x^3+1}{kx^2} dx = \frac{1}{k} \int x + \frac{1}{x^2} dx$ $= \frac{1}{k} \left(\frac{x^2}{2} - \frac{1}{x} \right) + C$ (b) $\frac{d}{dx} \ln(x^3+1) = \frac{3x^2}{x^3+1}$ ----- $\int_0^2 \frac{kx^2}{x^3+1} dx = \frac{k}{3} \int_0^2 \frac{3x^2}{x^3+1} dx$ $= \frac{k}{3} [\ln(x^3+1)]_0^2$ $= \frac{k}{3} \ln 9 \quad \left[\text{or } \frac{2k}{3} \ln 3 \right]$	
4	(i)  (ii) $\text{Area} = \int_{\frac{1}{2} \ln \frac{1}{2}}^0 2e^{2x} - 1 dx + \int_0^2 e^{-x} dx$ $= [e^{2x} - x]_{\frac{1}{2} \ln \frac{1}{2}}^0 + [-e^{-x}]_0^2$ $= 1 - \left(\frac{1}{2} - \frac{1}{2} \ln \frac{1}{2} \right) + (-e^{-2} + 1)$ $= \frac{1}{2} \left(3 + \ln \frac{1}{2} \right) - e^{-2} \quad \left[\text{or } \frac{1}{2} (3 - \ln 2) - e^{-2} \right]$	
5	(i)	

Method 1: $\begin{aligned} x^2 + 6x + 10 &= (x + 3)^2 - 9 + 10 \\ &= (x + 3)^2 + 1 \end{aligned}$ Since $(x + 3)^2 \geq 0$ for $x \in \mathbb{R}$, $(x + 3)^2 + 1 > 0$ for all real values of x (shown).	
Method 2: Since Discriminant $= 6^2 - 4(10) = -4 < 0$, and the coefficient of $x^2 > 0$, $x^2 + 6x + 10 > 0$ for all real values of x (shown).	
Method 3: $x^2 + 6x + 10 = (x + a)^2 + b = x^2 + 2ax + a^2 + b$ Comparing coefficients, $2a = 6 \Rightarrow a = 3$ $a^2 + b = 10 \Rightarrow b = 1$ Hence $x^2 + 6x + 10 = (x + 3)^2 + 1$. Since $(x + 3)^2 \geq 0$ for $x \in \mathbb{R}$, $(x + 3)^2 + 1 > 0$ for all real values of x (shown).	
(ii) $\frac{x^2 + 6x + 10}{x^2 + x - 6} > 0$ $\frac{x^2 + 6x + 10}{(x + 3)(x - 2)} > 0$ Since $x^2 + 6x + 10 > 0$ for all real values of x , $(x + 3)(x - 2) > 0$ $x < -3$ or $x > 2$ (iii) (a) $\frac{(\ln x)^2 + 6 \ln x + 10}{(\ln x)^2 + \ln x - 6} > 0$ Let $y = \ln x$ $\frac{y^2 + 6y + 10}{y^2 + y - 6} > 0$ $y < -3$ or $y > 2$ $\ln x < -3$ or $\ln x > 2$ $0 < x < e^{-3}$ or $x > e^2$	

	(b) $\frac{x^2 - 6x + 10}{x^2 - x - 6} > 0$ Let $y = -x \Rightarrow x = -y$ $\frac{y^2 + 6y + 10}{y^2 + y - 6} > 0$ $y < -3$ or $y > 2$ $-x < -3$ or $-x > 2$ $x > 3$ or $x < -2$	
6	(i) $y = \sqrt{9 - x^2}$ $\frac{dy}{dx} = \frac{-2x}{2\sqrt{9 - x^2}} = \frac{-x}{\sqrt{9 - x^2}}$ ----- When $x = 2$, $y = \sqrt{5}$ and $\frac{dy}{dx} = -\frac{2}{\sqrt{5}} = -\frac{2}{5}\sqrt{5}$. Hence equation of the tangent is $y - \sqrt{5} = -\frac{2}{5}\sqrt{5}(x - 2)$ ----- $y = -\frac{2}{5}\sqrt{5}(x - 2) + \sqrt{5}$ a = -2, b = 9 $= \frac{\sqrt{5}}{5}(-2x + 9)$ ----- Alternative method: When $x = 2$, $y = \sqrt{5}$ Gradient of line joining centre O and the tangent point = $\frac{\sqrt{5}}{2}$ Gradient of tangent line = $-\frac{2}{\sqrt{5}} = -\frac{2\sqrt{5}}{5}$ since a line tangent to a circle is perpendicular to the radius at tangent point. Hence equation of tangent line is $y - \sqrt{5} = -\frac{2}{5}\sqrt{5}(x - 2)$ ----- $y = -\frac{2}{5}\sqrt{5}(x - 2) + \sqrt{5}$ a = -2, b = 9 $= \frac{\sqrt{5}}{5}(-2x + 9)$ -----	

	(ii) Let $x = 0$, $y = \frac{9\sqrt{5}}{5}$. Therefore $B\left(0, \frac{9\sqrt{5}}{5}\right)$. ----- Let $y = 0$, $x = \frac{9}{2}$. Therefore $A\left(\frac{9}{2}, 0\right)$. ----- (iii) Area of shaded region = $\frac{1}{2}\left(\frac{9\sqrt{5}}{5}\right)\left(\frac{9}{2}\right) - \frac{1}{4}\pi(3)^2 = \frac{81\sqrt{5}}{20} - \frac{9}{4}\pi$. -----	
	--- End of Pure Maths Section --- Refer next page	

Section B: Statistics		
7	(i) A random sample of 100 applications means that each of the application in the population of 900 applications has an <u>equal chance</u> of being selected and are <u>independently</u> selected. (ii) First, obtain a list of all applications and assign sequential numbers to all of them. (Establish the sampling frame). Calculate the <u>sampling interval</u> $k = \frac{900}{100} = 9$. Choose a random number in the range 1 to 9 as a <u>starting point</u>. (e.g. we choose r as the starting value). Choose the r^{th}, $(r + 9)^{\text{th}}$, $(r + 18)^{\text{th}}$,, until 100 students are selected. <u>Alternative Presentation:</u> Sampling interval, $k = \frac{900}{100} = 9$ <u>List the applications in order of the applicants' names, and randomly select the first application within the first 9 applications to process.</u> Subsequently, <u>select every 9th application</u> until a sample of 100 applications is selected. (iii) A systematic sample may not ensure that all categories of scholarship applications were processed within that day if the <u>sampling interval</u> coincides with a <u>cyclic pattern</u> in the list of applications. <u>Alternatively:</u> A stratified sample ensures a <u>proportionate number of applications from each category</u> is being processed. (iv) <u>Random samples</u> of applications are chosen from each area of research with sample sizes <u>proportional to the size</u> of the respective categories. (using either simple random sampling or systematic sampling)	

8

(i)(a) When A and B are mutually exclusive, $P(A \cup B) = P(A) + P(B)$

$$P(B) = 0.8 - 0.4 = 0.4$$

(i)(b) When A and B are independent, $P(A \cap B) = P(A) P(B)$

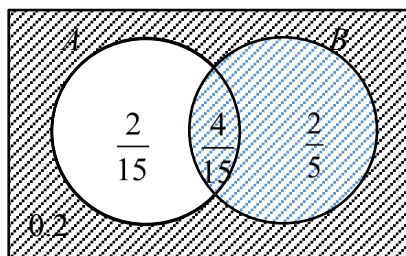
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$0.8 = 0.4 + P(B) - 0.4P(B) \quad \text{-----}$$

$$0.4 = 0.6 P(B)$$

$$P(B) = \frac{2}{3} \quad (\text{or } 0.667) \quad \text{-----}$$

(ii)



From Venn diagram above, $P(A' \cup B) = \frac{13}{15}$ (or 0.867)

Otherwise Method:

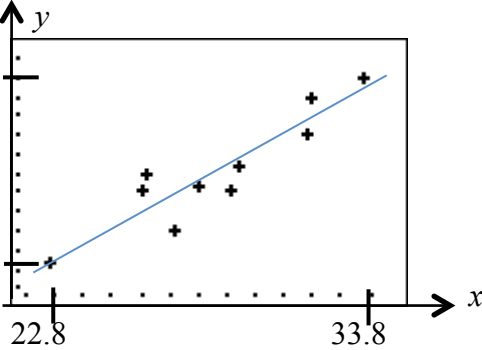
$$P(A' \cup B) = P(A') + P(B) - P(A' \cap B) = 0.6 + \frac{2}{3} - \left(\frac{2}{3} - 0.4 \left(\frac{2}{3} \right) \right) = \frac{13}{15}$$

(iii) Since C is a non-empty set, $P(C) > 0$.
and

$$\text{Since } C \text{ is a subset of } A \cap B', P(C) \leq 1 - P(A' \cup B) = 1 - \frac{13}{15} = \frac{2}{15}.$$

$$\therefore 0 < P(C) \leq \frac{2}{15}.$$

9	(i) Required probability = $(0.5)(0.05) + (0.35)(0.08) + (0.15)(0.13) = 0.0725$ (ii) $P(\text{A strawberry is produced by Orchard A} \mid \text{It is bruised})$ $= \frac{P(\text{A strawberry is produced by Orchard A} \cap \text{It is bruised})}{P(\text{A strawberry is bruised})}$ $= \frac{(0.5)(0.05)}{0.0725}$ ----- $= \frac{0.025}{0.0725}$ ----- $= 0.345 \text{ (3 s.f.)}$ ----- (iii) Required probability = $(0.9275)^3 + 3(0.9275)^2(0.0725) = 0.985 \text{ (3 s.f.)}$ Alternative Method: Let Y be the number of 'not bruised' strawberries out of 3 strawberries. $Y \sim B(3, 0.9275)$ $P(Y \geq 2) = 1 - P(Y \leq 1)$ $= 0.985 \text{ (3 s.f.)}$ (iv) Let X be the number of bruised strawberry in a box of 100 strawberries. $X \sim B(100, 0.0725)$ Since $n = 100$ is large, $np = 7.25 > 5$ and $n(1 - p) = 92.75 > 5$, $X \sim N(7.25, 6.724375)$ approximately. $P(X \geq 9) \stackrel{c.c}{=} P(X > 8.5) = 0.3148883854$ $= 0.315 \text{ (3 s.f.)}$ -----	
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10	(i)  (ii) $r = 0.91265$ (5 s.f.) = 0.913 (3 s.f.) The product moment correlation coefficient suggests a <u>strong positive linear correlation</u> between the <u>alkalinity of water</u> and the <u>hardness of water</u>. (iii) Equation of regression line of y on x: $y = 0.82x + 22.57$ (to 2 d.p.) (iv) When $x = 30$: $y = 0.82(30) + 22.57 = 47.17 = 47.2$ (1 d.p. following accuracy of data) Since the scatter diagram shows that the points lie close to a straight line and the <u>value of the correlation coefficient is very close to 1</u>, and the value $x = 30$ is within the data range, we are performing <u>interpolation</u> and the estimate is reliable. (v) The correct regression line would have the <u>same gradient</u> but the line would be <u>translated in the negative direction of the vertical axis by 3 units</u>.	
11	(i) Unbiased estimate of population mean, $\bar{x} = \frac{\sum (x - 760)}{100} + 760 = \frac{-150}{100} + 760 = 758.5.$ Unbiased estimate of population variance, $s^2 = \frac{1}{99} \left[3690 - \frac{(-150)^2}{100} \right] = 35.$ (ii) The term “unbiased estimate” means that the <u>expected value</u> of the <u>sample statistic</u> is <u>equal</u> to the <u>population parameter</u> that is being measured.	

- (iii) Let X be the length of a chinook salmon and μ be the population mean length of a chinook salmon.

To test $H_0 : \mu = 760$
 against $H_1 : \mu \neq 760$ at 5% level of significance.

Since $n = 100$ is large, by Central Limit Theorem:

$$\bar{X} \sim N\left(760, \frac{35}{100}\right) \text{ approximately under } H_0.$$

$$\text{Test statistic: } Z = \frac{\bar{X} - 760}{\sqrt{0.35}} \sim N(0,1).$$

$$z_{\text{test}} = \frac{758.5 - 760}{\sqrt{0.35}} = -2.5354628$$

Using GC, p -value = 0.0112 (3 s.f.).

```

Z-Test
Inpt:Data Stats
mu:760
sigma:5.9160797830...
x:758.5
n:100
mu:mu <mu >mu
Calculate Draw
    
```

```

Z-Test
mu:760
z:-2.535462764
p:=.0112299301
x=758.5
n=100
    
```

Since p -value = 0.0112 < 0.05, we reject H_0 and conclude that there is sufficient evidence at 5% significance level to support the biologist's claim that the mean length of a chinook is not 760 mm.

- (iv) To test $H_0 : \mu = 740$
 against $H_1 : \mu < 740$ at 1% level of significance.

Since $n = 60$ is large, by Central Limit Theorem:

$$\bar{X} \sim N\left(740, \frac{80}{60}\right) \text{ approximately under } H_0.$$

Since biologist's suspicion is not justified at 1% level of significance, we reject H_0 .

$$z_{\text{test}} = \frac{k - 740}{\sqrt{\frac{80}{60}}} < -2.3263 \quad (\text{Accept: } -2.326347877)$$

$$k < 737.31$$

Greatest possible value of $k = 737$ (nearest whole number).

12 (a)	X is normally distributed with mean 0. Given $P(X > -k) = p$, where $k > 0$: $P(X < -k) = 1 - p$. By symmetry of the normal distribution curve about mean 0: $P(X > k) = 1 - p$ $P(-k < X < k) = 1 - 2P(X < -k)$ $= 1 - 2(1 - p)$ $= 2p - 1$ <u>Alternative Method:</u> $P(X > -k) = p$ $P\left(Z > -\frac{k}{\sigma}\right) = p$ $P\left(Z < -\frac{k}{\sigma}\right) = 1 - p$ By symmetry of the normal distribution curve about mean 0: $P\left(Z > \frac{k}{\sigma}\right) = 1 - p$ $P(-k < X < k) = P\left(-\frac{k}{\sigma} < Z < \frac{k}{\sigma}\right)$ $= 1 - 2P\left(Z < -\frac{k}{\sigma}\right)$ $= 1 - 2(1 - p)$ $= 2p - 1$	
	<i>Refer next page.</i>	

12 (b)	(i) Let X be mass of a randomly chosen tablet of type A. $X : N(310, 10^2)$ Required probability $= P(X > 315) \times P(X < 305) \times 2$ $= 0.30853753^2 \times 2$ (Also accept: $0.30854^2 \times 2$) $= 0.1903908148$ (Also accept: 0.1903938632) $= 0.190$ (3 s.f.) ----- (ii) Let Y be mass of a randomly chosen tablet of type B. $Y : N(660, 30^2)$ Let $W = X_1 + L + X_6 - (Y_1 + Y_2 + Y_3) : W : N(-120, 3300)$ $P(-180 \leq W \leq 180)$ ----- $= 0.8518649534$ $= 0.852$ (3 s.f.) ----- (iii) Let T be mass of a randomly chosen box containing 10 tablets of type A. $T = X_1 + L + X_{10}$ $T : N(3100, 1000)$ ----- $P(T \leq W) = 0.90$ $P(Z \leq \frac{W - 3100}{\sqrt{1000}}) = 0.90$ ----- $W = 3140.52622$ $= 3141$ (nearest integer) ----- (iv) Let Q be number of batches out of 20 randomly chosen batches which will not meet the maximum weight limit. $Q : B(20, 0.10)$ $P(Q < 3) = P(Q \leq 2)$ $= 0.6769268$ $= 0.677$ (3 s.f.)	
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